

## \* Difference b/w Storage and Volume

### STORAGE

- \* The underlying physical or cloud storage resource
- \* Ex = SSD, HDD, SAN
- \* Provides the actual capacity where data is stored
- \* Can contain multiple Volumes

### VOLUME

- \* A logical unit of storage created from the underlying <sup>storage</sup> <sub>blocks</sub>
- \* Ex = disk partition, Kubernetes.
- \* Provides a way for app<sup>n</sup> or OS to access the storage.
- \* Usually uses part or all of the available storage.

\* Storage = The entire warehouse

\* Volume = A specific room or section inside the warehouse assigned for a particular purpose

\* Ex = Suppose you have a 1TB SSD:

\* Storage = The 1TB SSD itself.

\* Volume = You might divide it into:

• Volume A: 400 GB (OS)

• Volume B: 600 GB (Data)



② What is DNS & how DNS resolves addresses

→ Domain Name System is a system that translates domain names (like www.google.com) into IP addresses (like 142.250.190.78) so computers can find each other on the internet.

• Why do we need DNS?

→ Humans remember names easily, but computers communicate using IP addresses.

\* How DNS resolves addresses

① user enters a domain name

ex: www.google.com

② Browser cache check

③ OS cache check

④ DNS Resolver (Resolving Resolver) Browser cache

⑤ Root DNS server

⑥ Top-Level Domain

⑦ Authoritative DNS server

⑧ IP Address Returned

⑨ website opens

Flow Chart

↓ user

↓

↓ Browser cache

↓

↓ OS cache

↓

↓ DNS Resolver

↓

↓ Root DNS server

↓

↓ TLD Server (.com)

↓

↓ Authoritative DNS server

↓ Returns IP Address

↓ Browser connects to the website